

Appendix A

Setting Up Ubuntu

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Ubuntu is a user-friendly Linux distribution that provides a stable and powerful environment for programming, development, and system administration. Whether you're compiling code, learning Linux, or working with tools like gcc, git, and bash, having Ubuntu set up on your system is an essential first step.

This appendix will guide you through **three complete methods** for setting up Ubuntu:

1. **Native Installation (Dual Boot)** – Ubuntu runs directly on your hardware
2. **WSL (Windows Subsystem for Linux)** – Ubuntu runs inside Windows
3. **Virtual Machine (VM)** – Ubuntu runs inside a virtualization software

Each method is fully explained below so you can choose what suits your system and comfort level.



Where did Ubuntu come from? *The word Ubuntu comes from the Zulu and Xhosa languages of Southern Africa. It means “humanity toward others” or “I am because we are.” This philosophy of community and compassion is exactly what inspired South African entrepreneur Mark Shuttleworth to create Ubuntu in 2004.*

What started as a humble Linux project quickly grew into one of the world’s most user-friendly operating systems. Today, Ubuntu powers everything from personal laptops to cloud servers and even the International Space Station.

So when you’re setting up Ubuntu, you’re not just installing an OS—you’re joining a global tradition of open-source collaboration and shared learning. Pretty cool, right?

Method 1: Native Installation (Dual Boot Ubuntu + Windows)

Installing Ubuntu natively (i.e., directly on your system) is ideal when you want full performance, direct hardware access, and complete control. This guide walks you through a clean installation of **Ubuntu 20.04.1 LTS (64-bit)** using a bootable USB stick.

Note: Always download the **latest LTS version** from <https://ubuntu.com/download/desktop>. While this guide uses 20.04.1 as an example, newer LTS versions (e.g., 22.04.4) are available and preferred for security and hardware support.

Step 1: Back Up Your Computer

Before we do anything, we want to back up all files that are on your computer. This is an easy step, just use whatever your storage device of choice is. We're doing this because we will be messing around with the memory, so potentially something can get messed up.

Step 2: Download an Ubuntu

1. Visit <https://ubuntu.com/download/desktop>
2. Click Download under the latest LTS (Long-Term Support) release (e.g., Ubuntu 22.04 LTS or 20.04.6 LTS)
3. Save the `.iso` file (~4 GB) to your computer

☒ This `.iso` file will be used to create your bootable USB in the next step.

(Additional) Installing Ubuntu alongside Windows with BitLocker

If your Windows system drive is encrypted with **BitLocker**, follow these steps **before** proceeding with the installation:

1. Open **Control Panel** → **BitLocker Drive Encryption**
2. Locate your system drive (usually C:)
3. Click **Suspend protection** and confirm

After Ubuntu installation is complete, you can safely **re-enable BitLocker** from the same menu.

Important: If BitLocker is not suspended before installation, the Ubuntu installer may:

- Fail to resize the Windows partition
- Not detect your Windows system at all

To ensure a successful dual-boot setup, always **disable BitLocker protection temporarily** before launching the Ubuntu installer.

Step 3: Create a bootable USB stick

Use a tool like:

- Rufus (on Windows)
- Startup Disk Creator (on Ubuntu)
- Etcher (cross-platform)

Steps:

1. Insert a USB drive (8 GB or more).
2. Use the tool to select your downloaded .iso file.
3. Write it to the USB.
4. Safely eject when done.

Step 4: Boot from USB flash drive

1. Insert the USB into the computer where you want Ubuntu installed.
2. Reboot the machine.
3. Press the boot menu key (usually F2, F12, DEL, or ESC) to select USB.
4. From the Ubuntu welcome screen, click “Install Ubuntu”.

Step 5: Installation setup

2. Select your language and keyboard layout.
3. Optionally connect to Wi-Fi if prompted.
4. On the Updates and other software screen:
 - Choose Normal installation
 - Uncheck Download updates while installing
 - Uncheck Install third-party software (recommended to skip for now)

Click Continue.

Step 6: Type of installation

Ubuntu will detect any existing operating systems.

- Choose Install Ubuntu alongside Windows Boot Manager to dual-boot
- Avoid selecting Erase disk and install Ubuntu unless you're wiping the system
- Use Something else only if you're familiar with partitioning

Click Install Now, review the partition changes, and proceed.

Step 7: Create your login details

Enter the following:

- Your **name**
- Your **computer's name**
- A **username** (lowercase)
- A **password** (typed twice)

Choose whether to **require a password to log in**.

Click **Continue**.

Step 8: Choose your location

Use the map or drop-down to select your **region** or **time zone** (e.g., Asia → Karachi).

Click **Continue**.

Step 9: Ready to install

Ubuntu now installs:

- Progress will be shown with helpful welcome slides
- Installation time: ~15–25 minutes
- You may remove the USB stick after this step when prompted

Step 10: Complete the installation

Once installation finishes:

- Click **Restart Now**
- Remove the USB drive when prompted
- Your PC will reboot into Ubuntu (or show the **GRUB** boot menu if dual-booted with Windows)

Step 11: Don't forget to Update!

After login, open a terminal and run:

```
$ sudo apt update
$ sudo apt upgrade
```

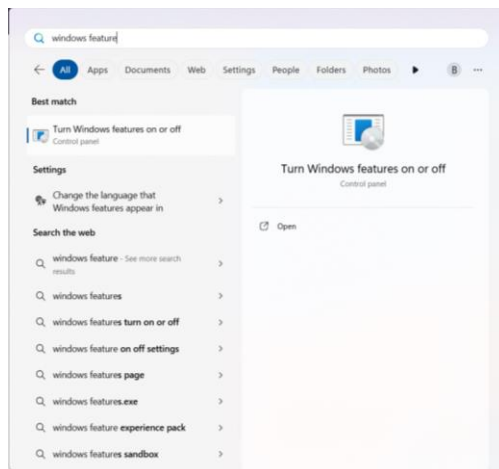
This fetches all the latest security and package updates.

Method 2: Setting Up Ubuntu via WSL

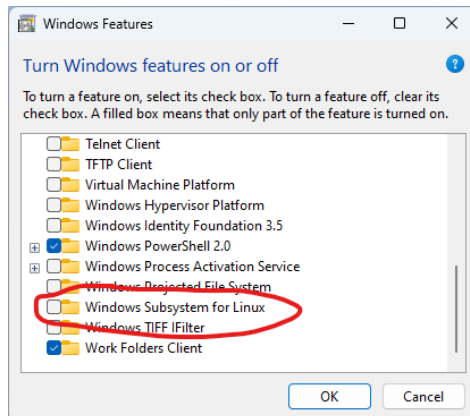
Want the power of Linux without leaving Windows? WSL makes it possible. No USBs. No dual booting. No BIOS drama. Just open a terminal and boom — you're running Ubuntu. It gives you a Linux terminal where you can install packages, compile code, and run shell scripts — all side by side with your Windows apps.

Step 1: Enable WSL in Windows Features

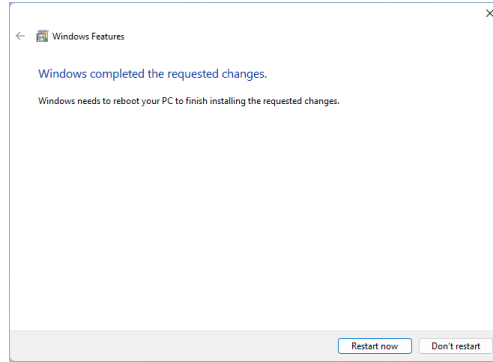
1. Go to your Windows search bar and type:



1. Click "Open" on the result that says "Turn Windows features on or off".
2. In the pop-up window, scroll down and check the box next to:



3. Click OK to begin the installation. A progress bar will appear as Windows enables the necessary components. After a short wait, you'll be prompted to restart your machine to complete the changes — go ahead and click “Restart Now”:

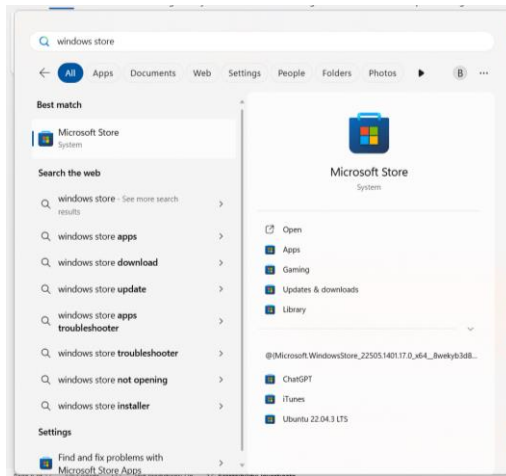


Your system will reboot with WSL enabled.

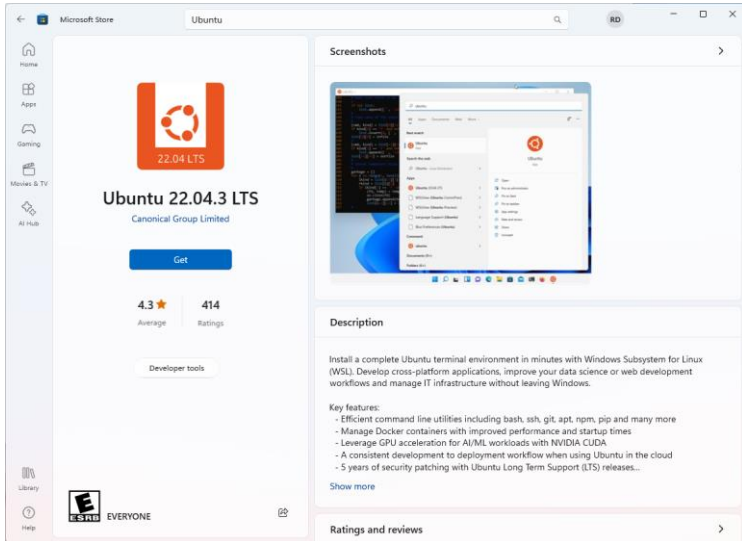
Step 2: Install Ubuntu from the Microsoft Store

Once you're back on your desktop:

1. Open the search bar again, type:



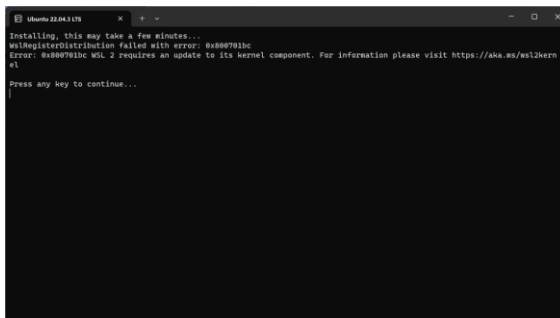
- Click “Open” to launch the store. In the Microsoft Store, search for:



- Click on whichever version is available.
- Click Get to download and install Ubuntu for WSL.

Step 3: Open Ubuntu and Handle First Launch Errors

After Ubuntu is installed, open the Start Menu. Type Ubuntu and click the app to launch it.



Possible Error: Kernel Update Required

If you see this error:

"The WSL 2 kernel is missing. Please visit <https://aka.ms/wsl2kernel> to install it."

Do this:

- Open your browser and go to: <https://aka.ms/wsl2kernel>
- Download the “WSL2 Linux kernel update package for x64 machines”
- Install it by clicking through the installer (Next → Finish)

Step 4: Virtualization Not Enabled?

If you see another error like:

"Please enable the Virtual Machine Platform Windows feature and ensure virtualization is enabled in the BIOS."

You have two possible fixes:

Fix A – From Windows Features:

1. Go back to **Windows Features**
2. Scroll down and enable:
☒ Virtual Machine Platform
3. Click **OK**
4. Reboot when prompted

Fix B – From BIOS/UEFI (Advanced):

- You may need to **enable virtualization** from your system's BIOS.
- Visit <https://aka.ms/enablevirtualization> for guidance based on your machine's brand.
- It usually involves entering BIOS during boot (DEL, F2, or ESC) and enabling **Intel VT-x / AMD-V**.

Step 5: Finish Ubuntu Setup

Once everything is working:

1. Launch Ubuntu again from the Start Menu
2. It will initialize and ask you to create:
 - A **UNIX username** (e.g., bilal)
 - A **password**

Once created, you'll land inside the Ubuntu shell prompt:

```
bilal@DESKTOP:~$
```


You now have a fully working Linux shell running inside Windows.

Step 6: Update Ubuntu

Run the following commands:

```
$ sudo apt update && sudo apt upgrade
```

Confirm any prompts with **Y** to install updates.

Access Your Windows Files from WSL

WSL makes your Windows files accessible via `/mnt`.

To access your `C:` drive, for example:

```
$ cd /mnt/c
```

To navigate to your Desktop:

```
$ cd /mnt/c/Users/<YourWindowsUsername>/Desktop
```

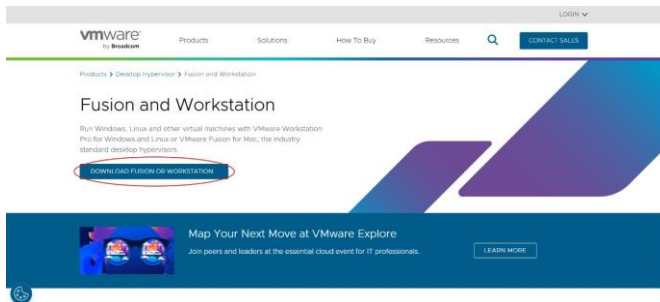
You can now read and write files from either side — compile `.c` files from Windows in Ubuntu, or edit Linux files using Windows editors.

Method 3: Installing Ubuntu via Virtual Machine

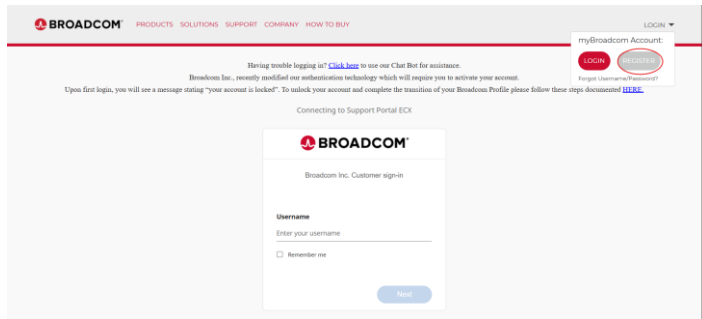
Virtual Machines (VMs) allow you to run Ubuntu on top of your existing Windows system without needing to dual-boot or modify partitions. VMware Workstation Pro is one of the most popular tools for this purpose. A VM behaves like a real computer with its own virtual CPU, memory, storage, and operating system — in this case, Ubuntu.

Step 1: Downloading VMware Workstation Pro

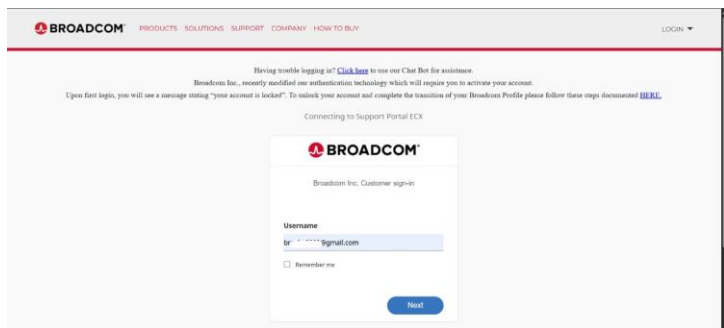
1. Search “vmware workstation pro” open the official page of VMware. Click download:



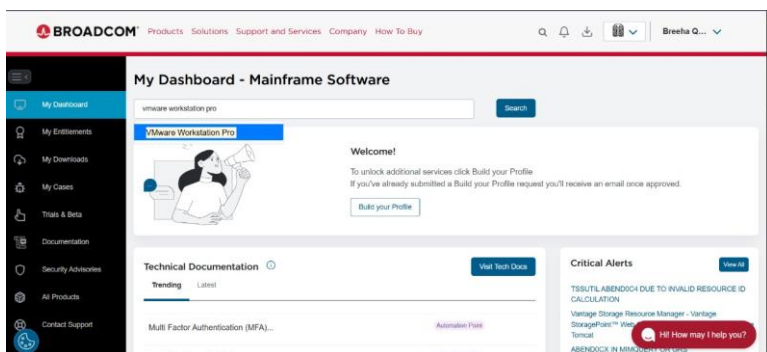
- It will redirect you to broadcom website. Now you'll have to create an account here:



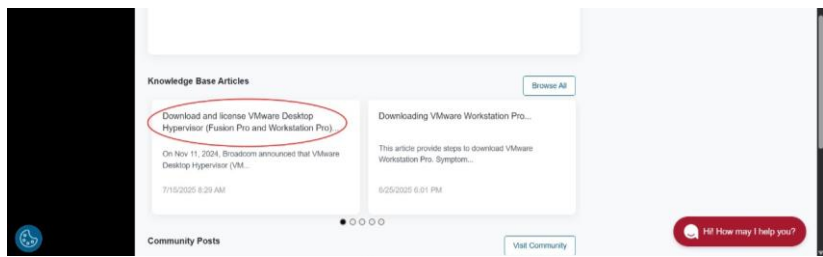
- Create an account by using any of your email, after doing User Registration. You'll have to Verify your Email Address by entering verification code. Then Complete your Registration by filling in your details.
- Now login to the website with your credentials:



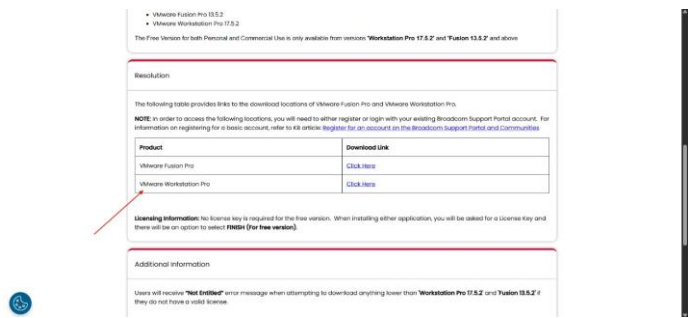
- Go to "My Dashboard" and type:



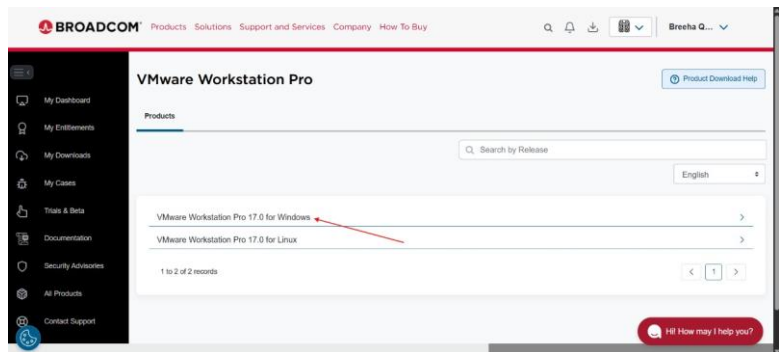
6. If it shows “No data found” then scroll down and click:

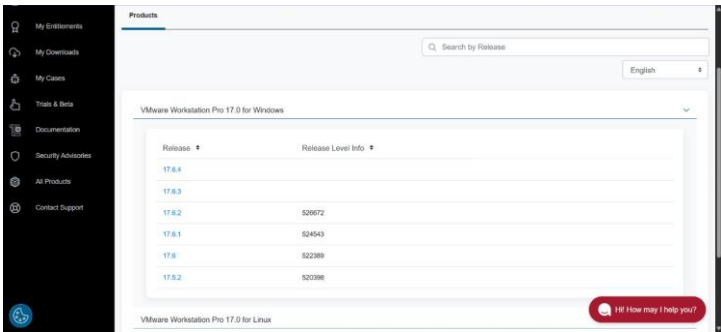


7. Download the free version

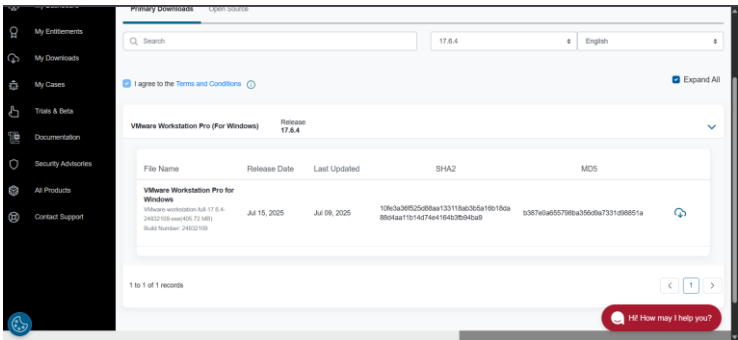


8. Now you'll see VMware Workstation Pro for different systems, download the latest version:



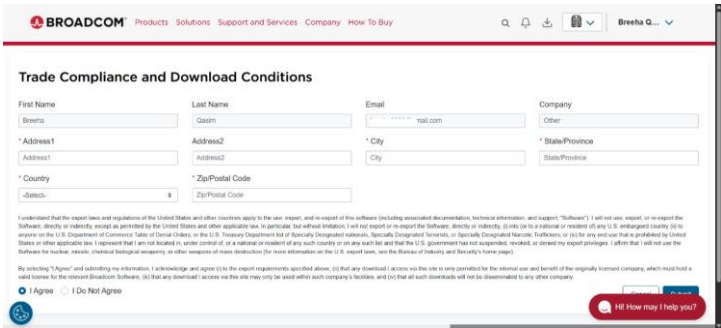


9. Read the “Terms and Conditions”, then click the checkbox only then you’ll be able to download:

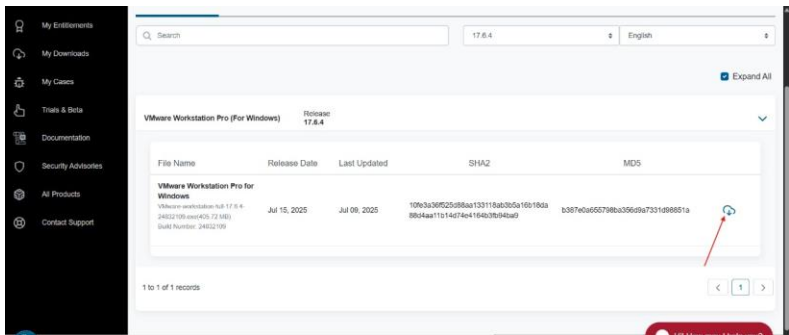


Click on “Yes”.

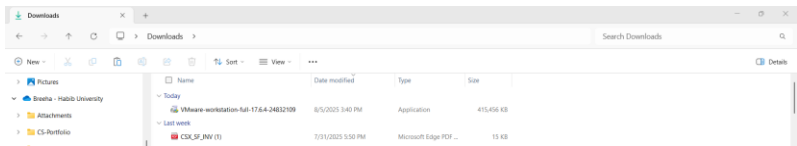
10. Now to download this you’ll have to provide address details:



11. After completing address verification process now click on download icon:



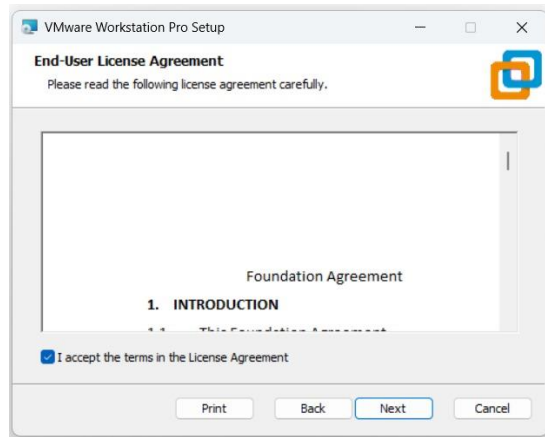
12. Click the VMware downloaded File and Install it.



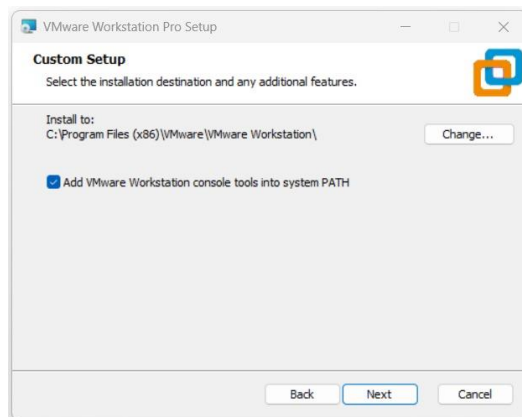
13. The **installation wizard** will launch. Click **Next** to begin setup.



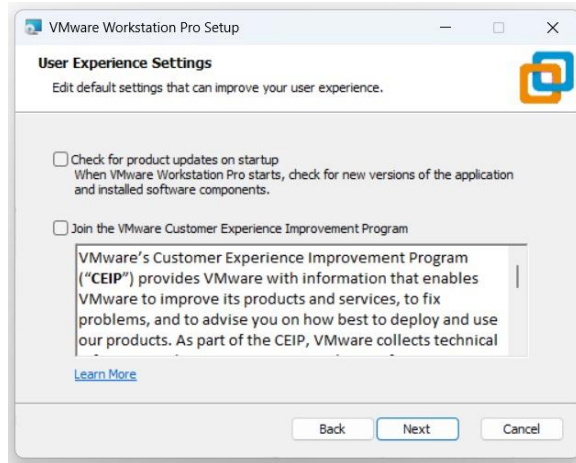
14. Accept the **VMware License Agreement** and click **Next**.



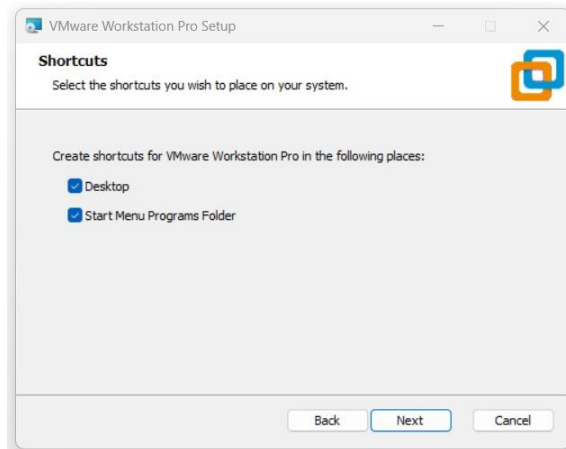
15. Choose the **installation location** or leave it as default. You may also enable the **Enhanced Keyboard Driver** here. Click **Next**.



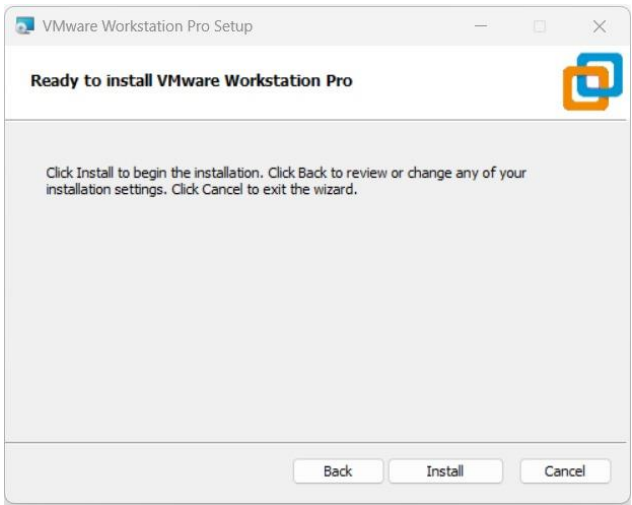
16. Disable both these options and click **Next**.



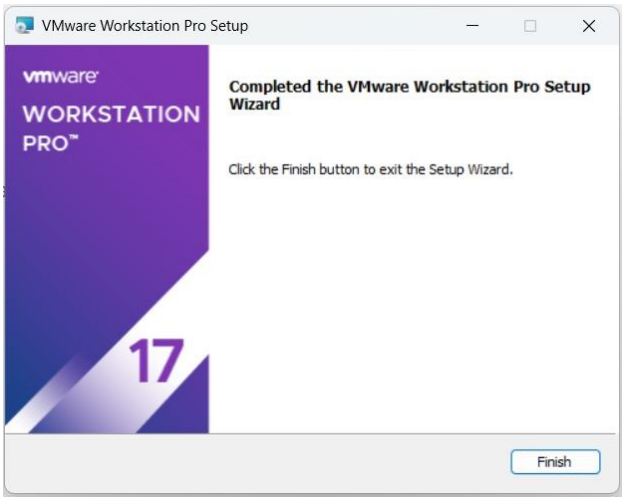
17. Select any **shortcuts** you want to create (Desktop, Start Menu). Click **Next**.



18. Click **Install** to begin installation.



19. Wait a few minutes. When the process finishes:

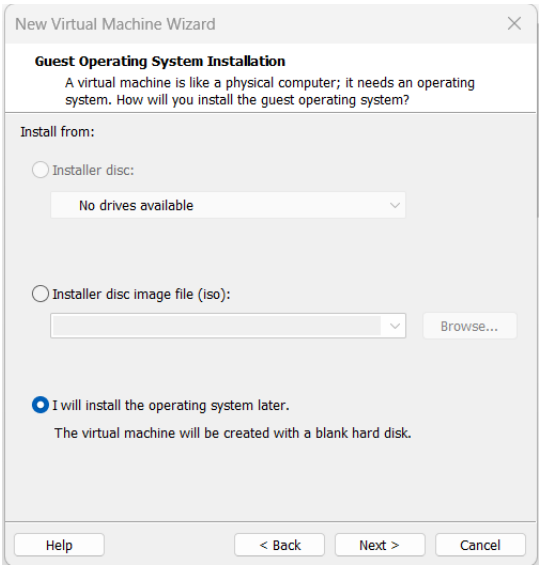


Step 2: Launch VMware and Prepare for Ubuntu

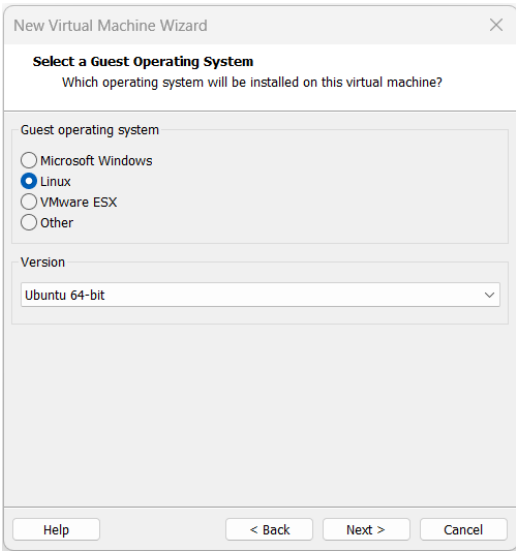
1. Visit the official Ubuntu download page: <https://ubuntu.com/download/desktop>
2. Download the latest Ubuntu Desktop LTS version (e.g., 22.04.3 LTS).
3. Save the .iso file in an easily accessible folder (like Downloads or Desktop).
4. Open VMware Workstation Pro
5. On the Home Screen, click “Create a New Virtual Machine.”
6. In the New Virtual Machine Wizard, select:



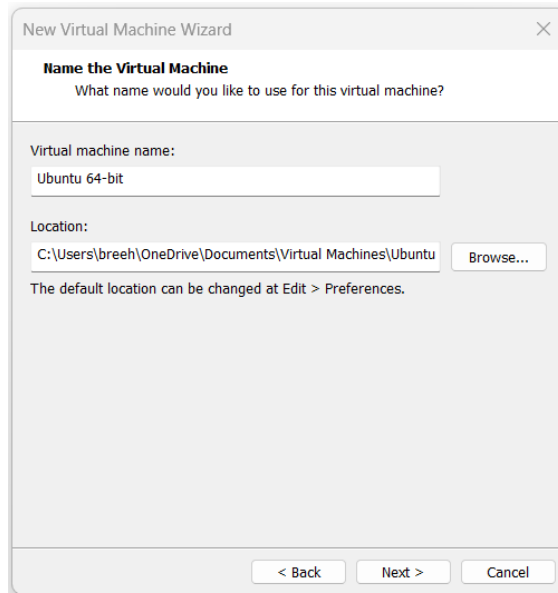
7. In the VMware wizard, select:



8. Select Guest Operating System: Linux and Version: Ubuntu 64bit



9. Enter a name for your virtual machine (e.g., **Ubuntu 22.04**) and choose a location to store the VM files → Click **Next**



New Virtual Machine Wizard

Name the Virtual Machine
What name would you like to use for this virtual machine?

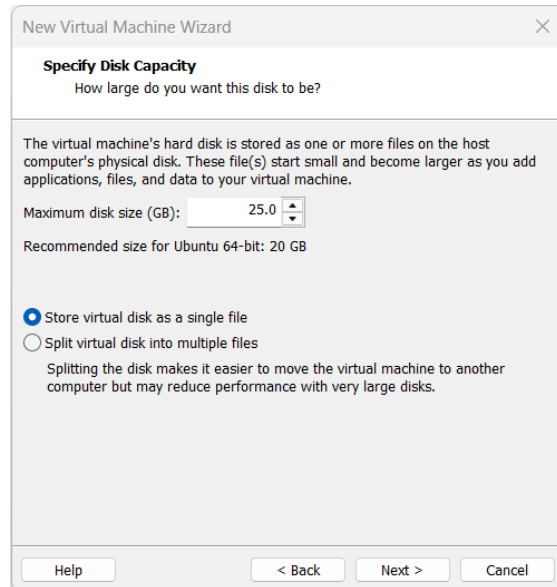
Virtual machine name:
Ubuntu 64-bit

Location:
C:\Users\breeh\OneDrive\Documents\Virtual Machines\Ubuntu Browse...

The default location can be changed at Edit > Preferences.

< Back Next > Cancel

10. Specify disk capacity:



New Virtual Machine Wizard

Specify Disk Capacity
How large do you want this disk to be?

The virtual machine's hard disk is stored as one or more files on the host computer's physical disk. These file(s) start small and become larger as you add applications, files, and data to your virtual machine.

Maximum disk size (GB): 25.0

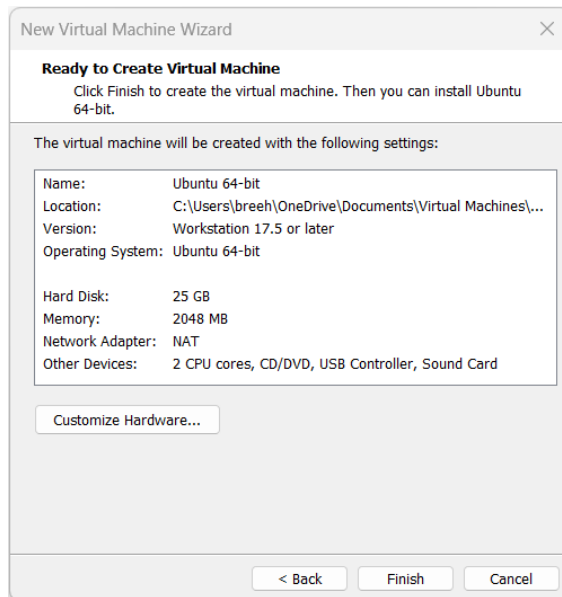
Recommended size for Ubuntu 64-bit: 20 GB

☒ Store virtual disk as a single file
☐ Split virtual disk into multiple files

Splitting the disk makes it easier to move the virtual machine to another computer but may reduce performance with very large disks.

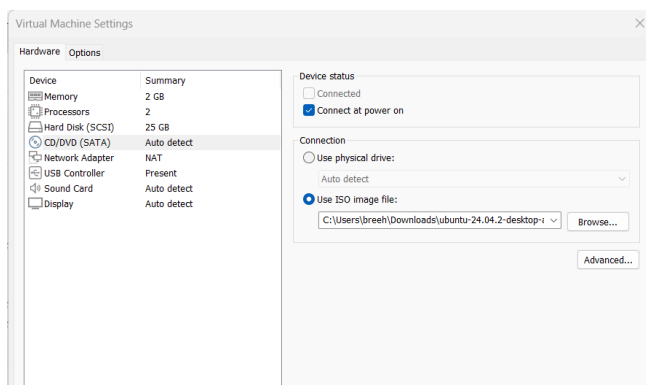
Help < Back Next > Cancel

- Review the summary and click Finish. Your VM is now created but has no operating system yet.



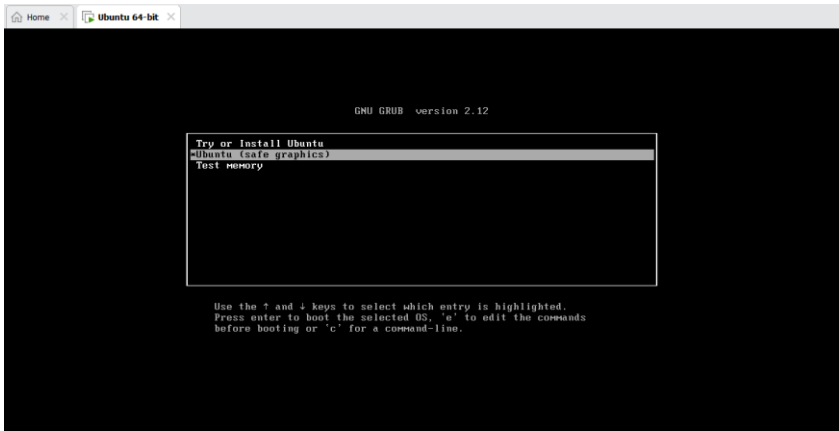
Step 3: Mount ISO and Install Ubuntu

- Select your newly created VM from the sidebar and click **Edit virtual machine settings**.
- Under Hardware, click CD/DVD (IDE).
- On the right, choose:



Then click Ok.

4. Now click Power on this virtual machine and select Ubuntu.



5. Follow the same installation process as described in **Method 1**
 - Select your language and keyboard layout
 - Choose Normal Installation
 - Set up partitions (you can use the entire virtual disk)
 - Set your username and password
 - Continue through time zone selection
 - Begin installation
6. Once the installer completes, restart the VM and eject the ISO if prompted.